Making Data

Downloading, Georeferencing, and Digitizing

ECON 895 Spatial Techniques in Empirical Economics

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Plan for tonight

Where data comes from

How georeferencing works

In QGIS

Analysis in R

Your paper

Tonight you need a browser and QGIS open. The break is at 8:45. We will start georeferencing before it.

The habits the exam grades

The lab exam is next Monday

Seventy minutes from 7:20, closed book and closed model, on Weeks 1–5 including tonight. The study guide is in `handouts/`.

Where data comes from

Everything you have used somebody made

- Dell (2010) built the *mita* boundary by matching a colonial list of *mita* districts to modern district polygons. That shapefile did not exist until she made it.

- Abramson (2017) traced the borders of hundreds of European states from historical maps, one period at a time.

- Nunn and Qian (2011) used a suitability surface that the FAO and IIASA built from soil, climate, and crop models. Tonight starts by fetching the current version of it.

The data you can download is the data everyone has

A language model can write every line of R from the past four weeks faster than you can. It cannot go to an archive or decide which of four thousand sheets your question needs. Novel data is what AI cheapens least and that is what sets a paper apart.

The potato surface from the FAO, step by step

1. Open `gaez.fao.org` and choose **Data Viewer**. Across the top, pick **4 – Suitability and Attainable Yield**.

2. In the **Filters** panel set **Sub-Theme Name** to Suitability Class and **Variable Name** to *Crop suitability index in classes; all land in grid cell*. Then **Time Period** 1981–2010, **Climate Model** CRUTS32, **Crop** White potato, **Water Supply** Rainfed, **Input Level** High.

3. Click anywhere on the map. The popup names the file, `siHr_wpo`, and carries a **Download URL**. That link is one GeoTIFF of 0.5 megabytes.

4. The **Table** tab under the map lists every raster the filters match, with a URL for each. **Export all as CSV** gives you the list.

Nunn and Qian (2011) used the FAO's 2002 release. The file you just found is version 4, published in 2021, and nothing on the page tells you the two differ.

The same download in R

```
url <- paste0("https://s3.eu-west-1.amazonaws.com/",
              "data.gaezdev.aws.fao.org/",
              "res05/CRUTS32/Hist/8110H/", "siHr_wpo.tif")
f <- "../data/wk05/siHr_wpo.tif"
if (!file.exists(f)) download.file(url, f, mode = "wb")

potato_v4 <- rast(f)
potato_v3 <- rast("../data/gaez_potato_suitability.tif")
compareGeom(potato_v4, potato_v3)
```

```
## [1] TRUE
```

The URL is the filter panel written out. Bucket, module `res05`, climate source, period and input level `8110H`, then the variable and crop. Change `wpo` to `whe` and you have wheat. The file lands on the same grid as the one Week 4 used, and `compareGeom` says so.

New numbers on the same grid

```
freq(potato_v4)[1:3, ]
```

```
##   layer value   count
## 1     1     0 7048266
## 2     1     1   52515
## 3     1     2   94418
```

```
global(potato_v4, "mean")
```

```
##                mean
## siHr_wpo 1.665997
```

```
global(classify(potato_v4, cbind(0, NA)), "mean", na.rm = TRUE)
```

```
##                mean
## siHr_wpo 6.809549
```

Zero is a value

Week 4's file stored the ocean as NA. This one stores it as 0, and a mean will use it. The portal's legend says 0 is no data. The file does not.

Across the version change 74 percent of land cells keep their class, and at Week 4's 1,797 cities the share on class 4 or better moves from 17 to 58 percent.

Four crops as one object

```
crops <- c(potato = "wpo", wheat = "whe", barley = "brl", maize = "mze")
files <- paste0("../data/wk05/siHr_", crops, ".tif")

suit <- rast(files); names(suit) <- names(crops)
dim(suit)                                  # rows, columns, layers
```

```
## [1] 2160 4320    4
```

```
names(suit)
```

```
## [1] "potato" "wheat"  "barley" "maize"
```

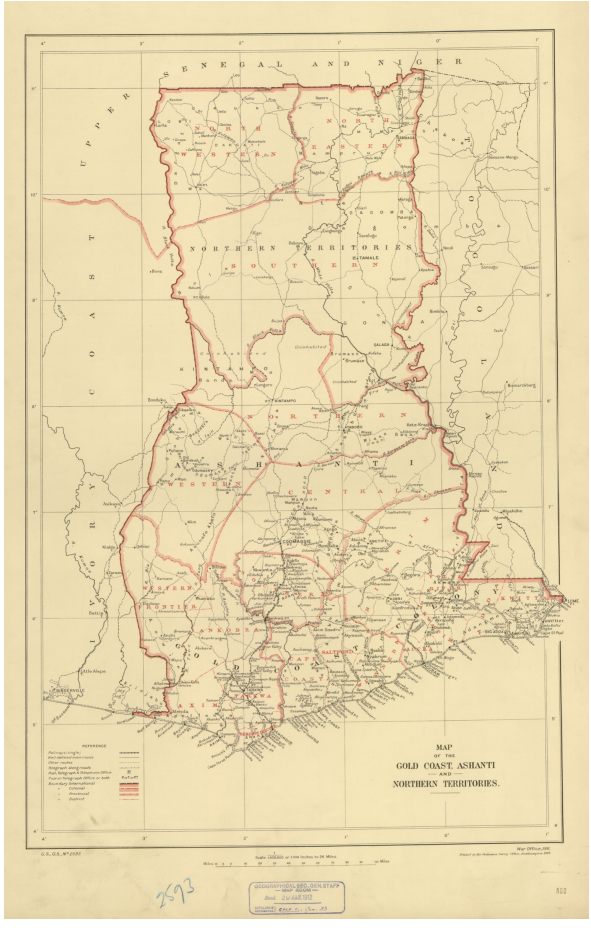
```
at_city <- terra::extract(suit, vect(pts))[, -1]
sapply(at_city, function(v) mean(v %in% 1:4))
```

```
##    potato     wheat    barley     maize
## 0.5843072 0.6850306 0.6694491 0.2910406
```

`rast` stacks the four files because their grids match, and it would refuse if they did not. One `extract` then returns a column per layer. The other-crop suitability controls in Nunn and Qian (2011) are one call.

How georeferencing works

Tonight's second object



A 1911 survey map of the Gold Coast, Ashanti, and the Northern Territories. Nineteen megabytes of scanned paper.

The FAO file arrived with a grid and a CRS. This one arrived with neither. Nothing in it can enter a regression. By the end of class it will be a dataset.

What a scan is missing

A raster needs an extent and a CRS. A scan has neither, so every pixel is addressed by row and column and nothing else.

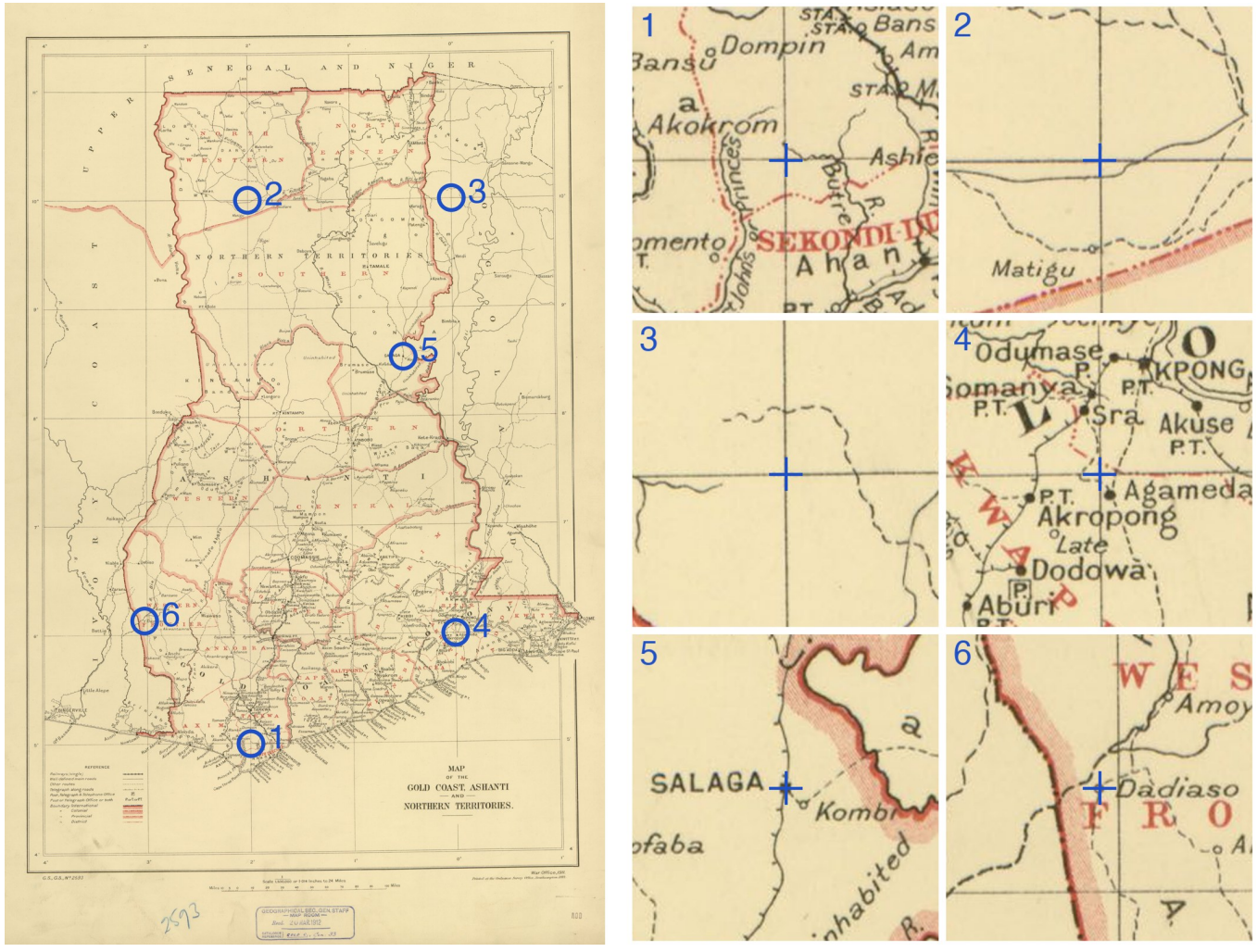
A ground control point is one observation of the missing function

It is a place where you know both the pixel address and the real-world coordinate. Collect a few and you can estimate the function that maps one to the other.

- Many printed maps carry a graticule, a grid of latitude and longitude lines. Where two lines cross, the map states the coordinate itself.

- Without a graticule you match features such as towns, river confluences, or church towers to a modern source. Each point is then only as good as that match.

The six control points on the 1911 map



Points 1–4 sit where graticule lines cross, at 5°N 2°W, 10°N 2°W, 10°N 0°, and 6°N 0°. The map prints those coordinates. Point 5 is the town of Salaga and point 6 is Dadiaso on the western frontier. Their coordinates had to come from outside the map.

Georeferencing is a regression

World coordinate on the left, pixel position on the right.

```
fit1 <- lm(cbind(mapX, mapY) ~ pixelX + pixelY, data = gcp)

sqrt(mean(rowSums(residuals(fit1)^2)))   # RMS miss per point, degrees
```

```
## [1] 0.01440022
```

That is the order-1, or affine, transformation. Its six parameters cover translation, rotation, scale, and shear. Order 2 adds quadratic terms for twelve parameters in all. A thin-plate spline bends the sheet until it passes through every point exactly.

QGIS calls this a **transformation type**. It is a model, and you are choosing its flexibility.

Two regressions in one call

With `cbind(mapX, mapY)` on the left, `lm` fits two equations on the same regressors.

$\text{mapX}_i = a_0 + a_1\,\text{pixelX}_i + a_2\,\text{pixelY}_i + u_i \qquad \text{mapY}_i = b_0 + b_1\,\text{pixelX}_i + b_2\,\text{pixelY}_i + v_i$

```
coef(fit1)                                  # one column per equation
```

```
##                       mapX          mapY
## (Intercept) -4.353132e+00  1.183058e+01
## pixelX       1.734806e-03 -1.199562e-05
## pixelY       1.266794e-05  1.641261e-03
```

```
coef(lm(mapX ~ pixelX + pixelY, data = gcp)) # the first column, alone
```

```
##   (Intercept)        pixelX        pixelY
## -4.353132e+00  1.734806e-03  1.266794e-05
```

Each column is ordinary OLS. When every equation has the same regressors, estimating them jointly gives the same coefficients as running them one at a time. The three coefficients in each column make the six parameters of the affine transformation.

The two equations meet only when we score the fit. `rowSums` adds each point's miss in longitude and in latitude, so the RMS is a distance per point.

Does a more flexible transformation fit better?

As you place control points, QGIS reports a residual for each one, the distance between where you clicked and where the fitted transformation puts that point. Small residuals look like a good georeference. Order 2 adds squared and interaction terms, six coefficients per equation instead of three.

```
fit2 <- lm(cbind(mapX, mapY) ~ poly(pixelX, 2, raw = TRUE) +
             poly(pixelY, 2, raw = TRUE) + pixelX:pixelY, data = gcp)

c(order1 = sqrt(mean(rowSums(residuals(fit1)^2))),
  order2 = sqrt(mean(rowSums(residuals(fit2)^2))))
```

```
##     order1     order2
## 0.01440022 0.00000000
```

A zero residual is arithmetic

Six points give twelve numbers to match, and order 2 has twelve coefficients. It passes through every point because it has no choice. The residual measures fit at the points you chose, and your data lives in the space between them.

A residual cannot tell you how accurate the map is. The next slide measures that.

Hold a point out

```
loo <- sapply(seq_len(nrow(gcp)), function(i) {
  m <- lm(cbind(mapX, mapY) ~ pixelX + pixelY, data = gcp[-i, ])
  p <- predict(m, newdata = gcp[i, ])
  sqrt(sum((as.numeric(p) - as.numeric(gcp[i, c("mapX", "mapY")]))^2))
})
c(in_sample = sqrt(mean(rowSums(residuals(fit1)^2))) * 111,
  held_out  = sqrt(mean(loo^2)) * 111)   # km, at about 111 km a degree
```

```
## in_sample  held_out
##  1.598425  2.745353
```

Fit on five and predict the sixth, six times over. The RMS miss is **2.7 kilometers** held out, against 1.6 in sample. The held-out figure is this map's **accuracy budget**, and every measurement you take from the map inherits it.

All six points lie between about 3°W and the prime meridian and between 5°N and 10°N. The north, the west, and the strip east of the meridian are extrapolation, and the budget there is larger.

Order 2 on the same test comes out about the same, with one point missed by five kilometers. Six points cannot tell the two apart. More points can.

What the accuracy budget tells you

Read any location off the warped map and expect it to sit about 2.7 kilometers from where the place really is, in some direction. The US federal standard for map accuracy (NSSDA, 1998) turns an RMS miss into a 95 percent radius by multiplying by 1.73. Here that is about 5 kilometers.

- Every later step inherits it. A town's coordinates, the side of a boundary a unit falls on, and a distance-to-railway treatment all carry error of this size, and classical error in a treatment attenuates its estimate.

- It is itself noisy. The six held-out misses run from 1.0 to 3.9 kilometers, so say "about three kilometers."

- It holds inside the control points. Outside them the fit extrapolates and the error grows.

- It describes absolute position. An affine fit shifts nearby points together, so the distance between two nearby towns is more accurate than either town's coordinates.

- It leaves out digitizing. Your clicks when you trace a feature add their own error.

In QGIS

The Georeferencer step by step

1. **Layer** then **Georeferencer**. A second window opens with its own menu bar and toolbar.

2. **Open Raster** and choose `data/wk05/gold_coast_1911.png`. It appears with pixel coordinates along the edges.

3. Zoom to a graticule intersection and click **Add Point**, then click the crossing. The **Enter Map Coordinates** dialog opens. Set its CRS to EPSG:4326 before typing. Longitude is X and latitude is Y, and west and south are negative.

4. Place six and spread them across the sheet. Points clustered in one region tell you nothing about the rest. Watch the GCP table fill in under the image.

If you fall behind, choose **File** then **Load GCP Points** and open `gold_coast_1911.points`, which holds the set I placed.

Transformation settings and the reason for each

1. **Transformation Settings**, the gear on the toolbar. **Transformation type:** Polynomial 1. The **Linear** option above it has no rotation or shear. Polynomial 2 needs at least six points, and with exactly six it fits them perfectly.

2. **Resampling method:** Nearest neighbour. It keeps every pixel value as scanned.

3. **Target CRS:** EPSG:4326. The 1911 datum is undocumented, so this is a relabel in the Week 1 sense. Datum shifts of that era run to a few hundred meters and sit inside the budget.

4. **Output raster:** `data/wk05/gold_coast_1911_georef.tif`. The default appends `_modified`, and the R half reads the name above. Leave **Load in project when done** checked.

5. **Start Georeferencing**. Before you close the window, read the residual column we just spent twenty minutes learning to distrust.

If the Georeferencer misbehaves

Each row is a symptom you may see and what to do about it.

What you see	What to do
The point lands far from where you clicked.	The coordinate dialog is using the project CRS. Set it to EPSG:4326 before typing.
The residuals look enormous.	They are in pixels. **Settings**, **Configure Georeferencer**, **Residual units** switches to map units.
R cannot find your raster.	QGIS added `_modified` to the name. Rename the file or type the name in the settings.
The map comes out mirrored left to right.	A longitude lost its minus sign. West is negative, so check every X in the GCP table.

Break

Ten minutes. Leave the warp running if it has not finished.

Digitizing

1. Close the Georeferencer and work in the main window on the warped GeoTIFF.

2. **Layer**, **Create Layer**, **New GeoPackage Layer**. Save it as `data/wk05/my_rail.gpkg` with geometry type **Line**, CRS EPSG:4326, and a text field called `name`. **Add to Fields List**, then **OK**.

3. **Toggle Editing**, the pencil, then **Add Line Feature**. Trace the railway from Sekondi on the coast north through Tarkwa to Kumasi, which the sheet spells Coomassie. Left-click adds a vertex and right-click ends the line. A form asks for `name`.

4. **Save Layer Edits** before you toggle editing off.

Save to GeoPackage

A GeoPackage is one file. It keeps long field names, though sf names its geometry column `geom`, and it stores its CRS inside the file. A shapefile needs at least three files plus a `.prj` for its CRS.

If digitizing misbehaves

Each row is a symptom you may see and what to do about it.

What you see	What to do
The line ends after one click.	Left-click adds a vertex and right-click finishes the line.
A form pops up after every feature.	Fill in `name` and click **OK**, or turn the form off under **Settings**, **Options**, **Digitizing**.
Vertices do not meet where lines join.	**Enable Snapping**, or press S, so new vertices snap to existing ones.
Toggling editing off asks **Save**, **Discard**, or **Cancel**.	Choose **Save**. **Discard** throws away everything you traced since the last save.

Analysis in R

The warped raster in R

```
sheet <- rast("../data/wk05/gold_coast_1911_georef.tif")

c(rows = nrow(sheet), cols = ncol(sheet), bands = nlyr(sheet))
```

```
##  rows  cols bands
##  5224  3547     4
```

```
crs(sheet, describe = TRUE)$name
```

```
## [1] "WGS 84"
```

```
round(as.vector(ext(sheet)), 2)
```

```
##  xmin  xmax  ymin  ymax
## -4.42  1.50  3.12 11.83
```

If you saved your warp under that name, this is your raster. It has an extent and a CRS because you gave it both. A picture became a raster the moment somebody supplied six control points.

run it

Does it land where it should?



The heavy red line is the modern Ghana boundary. The coast and the western frontier agree with the sheet. In the east the red line encloses British Togoland, which was German in 1911 and joined Ghana at independence in 1957. The border moved, and the map is accurate to about 2.7 kilometers.

run it

The Sekondi–Kumasi railway, traced



One line in 67 clicks, from Sekondi on the coast through Tarkwa, Dunkwa, and Obuasi to Kumasi, which the sheet spells Coomassie. It opened in 1903.

run it

Measuring the trace

```
rail <- st_read("../data/wk05/sekondi_kumasi_trace.gpkg", quiet = TRUE)

round(c(km       = as.numeric(st_length(rail)) / 1000,
        vertices = nrow(st_coordinates(rail))), 1)
```

```
##       km vertices
##    232.8     67.0
```

233 kilometers from 67 clicks. The line as built runs 168 miles, or 269 kilometers, so the trace comes up 36 kilometers short.

`st_length` returns meters on EPSG:4326 because sf measures on the sphere.

run it

Length under the accuracy budget

- A shift moves every vertex together and leaves the length alone. The budget says where the line is.
- Length error comes from the clicks. A trace cuts every curve it does not follow, so it runs short, and more clicks move it toward the true length.
- Measure your own trace, then compare with the person next to you. The spread in the room is the digitizing error.

```
sum(st_length(st_read("../data/wk05/my_rail.gpkg"))) / 1000
```

Where the budget bites

Put a five-kilometer buffer around this line and its edge is uncertain by about half the buffer's width. Anywhere along the route a unit within 2.7 kilometers of that edge could sit on either side of it.

Shapefiles abbreviate field names

```
demo <- st_sf(
  a_very_long_field_name = 1:2,
  another_long_one = c("x", "y"),
  geometry = st_sfc(st_point(c(0, 0)), st_point(c(1, 1)),
                    crs = 4326))
f <- tempfile(fileext = ".shp");  st_write(demo, f, quiet = TRUE)
```

```
## Warning in abbreviate_shapefile_names(obj): Field names abbreviated
     for ESRI Shapefile driver
```

```
g <- tempfile(fileext = ".gpkg"); st_write(demo, g, quiet = TRUE)

names(st_read(f, quiet = TRUE))
```

```
## [1] "a_vr___"  "anthr__"  "geometry"
```

```
names(st_read(g, quiet = TRUE))
```

```
## [1] "a_very_long_field_name" "another_long_one"
## [3] "geom"
```

The attribute table is dBASE, which limits field names to ten characters. sf abbreviates with one warning that scrolls past, and QGIS truncates to ten. Either way the column is renamed for good.

Your paper

What map would your paper digitize?

The question to leave with

What sheet, census table, or register exists on paper and has never been made machine readable?

- Scans live in the David Rumsey Map Collection, the Library of Congress, national and colonial archives, and the map room of almost every research library.
- The best sources sit unscanned in archives, and finding them takes a trip.
- An evening of georeferencing and digitizing produces one railway. A term produces a paper's worth, and everything from the past month applies to it the moment it is data.

The evening in one slide

1. A downloaded file carries a version. Record it and check what its zeros mean.

2. Files on one grid stack into one object, and one `extract` reads them all.

3. A control point is an observation. The transformation is a model and you choose its flexibility.

4. A saturated fit reports zero error. Hold a point out and report that number instead. Ours was 2.7 kilometers, and it bounds where features sit.

5. Count your features and then look at each one.

Before next week

The lab exam is Monday, 7:20–8:30

Seventy minutes, closed book and closed model, on Weeks 1–5. It asks you to build and to check. You will not operate QGIS. From tonight it asks what a georeferencing residual reports and what it stays silent about.

1. Read `handouts/lab_exam_study_guide.pdf`.

2. On Sunday run `git pull` and then `code/00_setup_test.R`, and confirm every check passes.

3. Work the weekly scripts by hand with the slides closed.

4. Start a list of maps your own paper could be built on.

After the exam Week 6 turns to language models that read sources for you.

References

Abramson, S. F. (2017). The economic origins of the territorial state. *International Organization*, 71(1):97–130.

Dell, M. (2010). The persistent effects of Peru's mining mita. *Econometrica*, 78(6):1863–1903.

Nunn, N. and Qian, N. (2011). The potato's contribution to population and urbanization: Evidence from a historical experiment. *The Quarterly Journal of Economics*, 126(2):593–650.